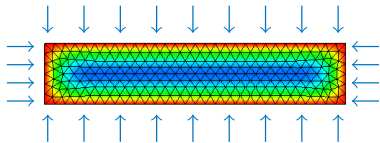


Example – Aging of fiber-reinforced polymers

How can we model this? Let us try a **finite element model** with water and mechanics:

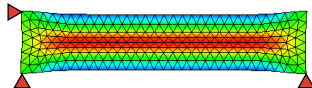
Water diffusion:

$$\frac{\partial c}{\partial t} - D \nabla \cdot (\nabla c) = 0$$



Stress equilibrium:

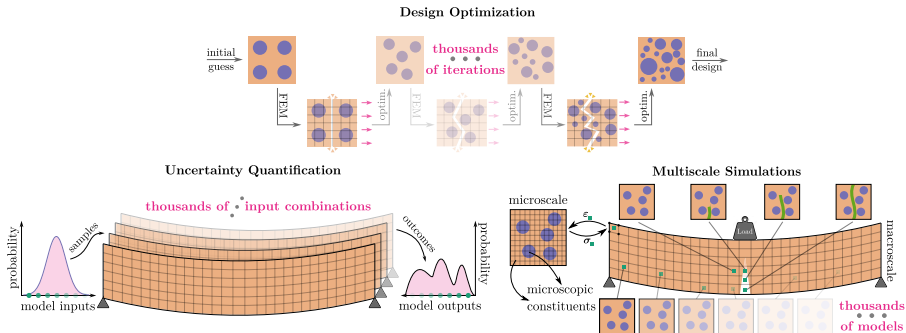
$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0}$$



Good models do not come cheap

We need a way to run these models much faster:

- Highly detailed models **can take months to run** and we need to run a lot of them

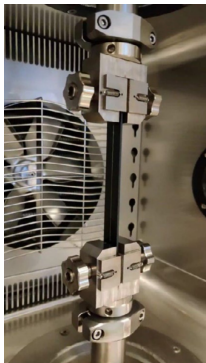


What we are now able to do: Reproducing real experiments

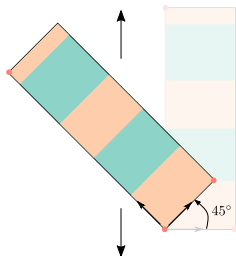
We also used it to confirm some experimental results that had been puzzling us:

- Uniaxial tension on off-axis carbon-PEEK coupons

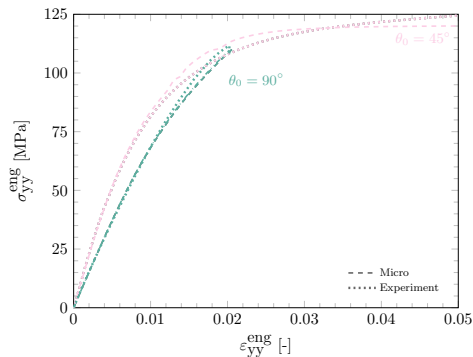
Uniaxial tension experiment



Off-axis angle

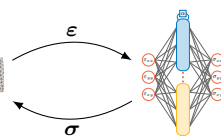
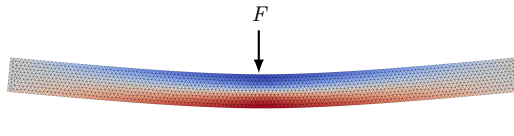
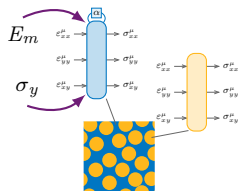


Engineering stress-strain



[Maia, Rocha, van der Meer (2025), [arXiv:2501:10193](#)]

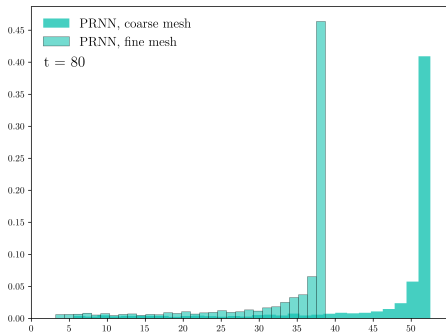
What we are now able to do: Multiscale UQ



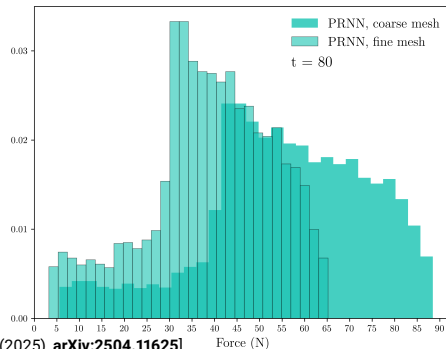
Three-point bending, **fine** macro mesh:

- Monte Carlo forward UQ, $N = 5000$ samples
- PRNN trained with a single (E_m, σ_y) pair
- Sequential FE² UQ would now take **≈ 50 years to run**

$$p(F) = \int p(F|E_m)p(E_m) dE_m$$



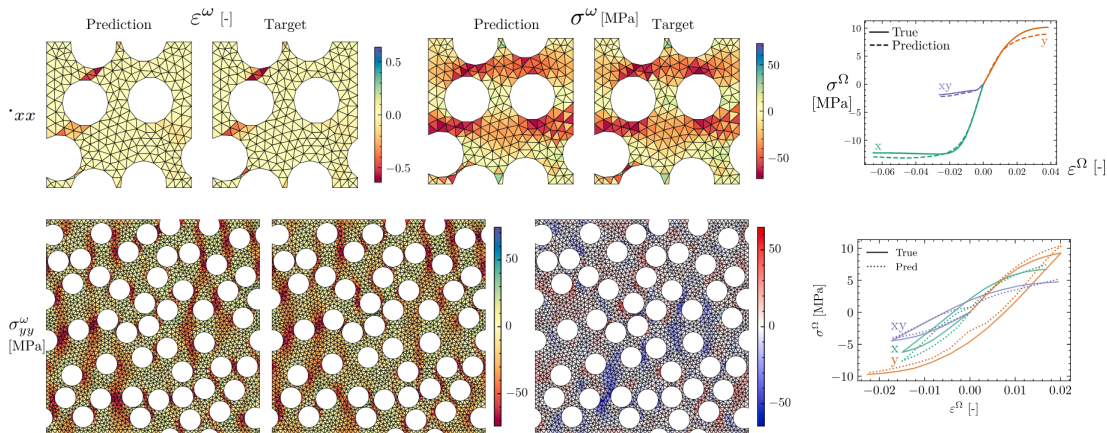
$$p(F) = \int p(F|E_m, \sigma_y)p(E_m, \sigma_y) dE_m d\sigma_y$$



A GNN for multiscale solid mechanics

We basically have an encode-process-decode-material approach:

- GNN computes strains (ε), material models compute stresses (σ)
- We train on models with up to 9 circular features, transfer to up to 64 features without retraining



[Storm, Rocha, van der Meer (2024), **CMAME**]